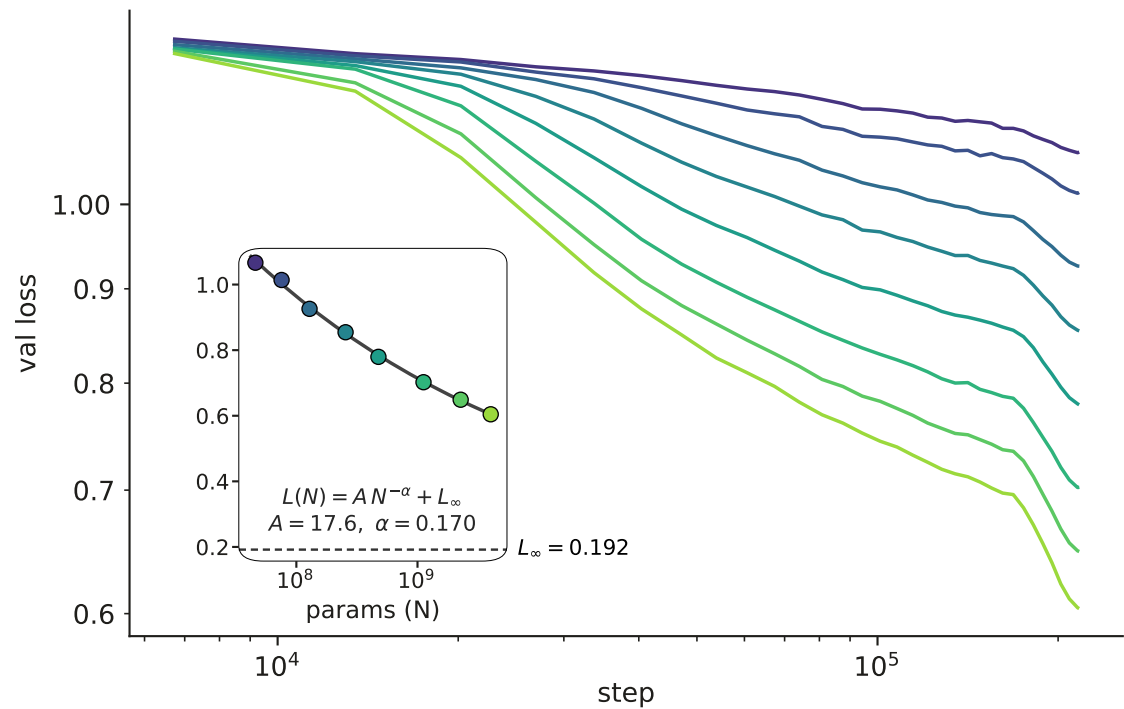
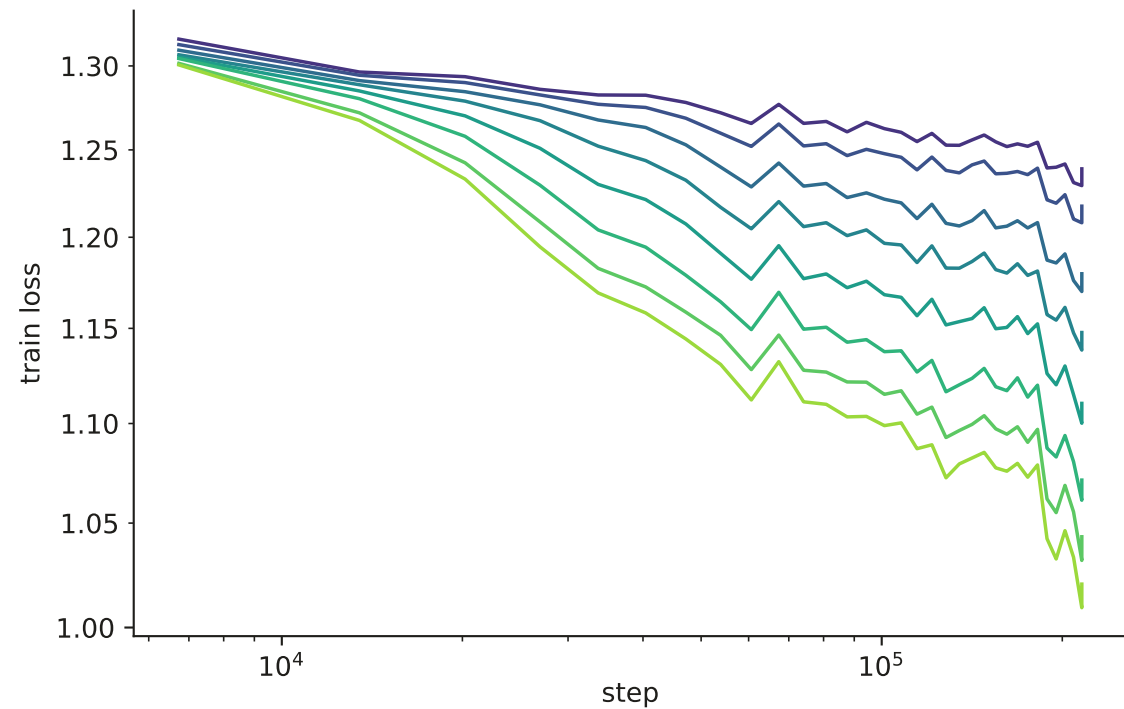


Parameter scaling — loss curves & scaling law
 $N = 46\text{M} - 4\text{B}$, $D = 84\text{B}$, $C = 2.5 \times 10^{19} - 2.1 \times 10^{21}$



model params
46M 76M 128M 255M 476M 1B 2B 4B